BEING GORENSTEIN IS A PROPERTY OF THE FORMAL FIBRES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The permanence of the Gorenstein property under a local complete intersection extension. Invoked inside the counted proof (source lines 4427-4428). Carries the property through the relative global complete intersection used in part (A).
- The preservation of the Gorenstein property under a flat local homomorphism with Gorenstein fibre. Invoked inside the counted proof (source lines 4443-4444 and 4447). Gives parts (C) and (D).
- The Gorenstein property of a regular ring. Invoked inside the counted proof (source line 4442). Supplies the Gorenstein fibre in the regular case.

Source reference. Original reference algebra-lemma-colimit-syntomic: sources/stacks-dualizing/pinned-original-source.tex, source lines 4426–4426. Navigation pointer only: the counted text refers to the Stacks item algebra-lemma-colimit-syntomic, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-flat-under-gorenstein: sources/stacks-dualizing/pinned-original-source.tex, source lines 4164–4164. Navigation pointer only: the counted text refers to the Stacks item lemma-flat-under-gorenstein, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-gorenstein-lci: sources/stacks-dualizing/pinned-original-source.tex, source lines 4136–4136. Navigation pointer only: the counted

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text refers to the Stacks item lemma-gorenstein-lci, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-regular-gorenstein: sources/stacks-dualizing/pinned-original-source.tex, source lines 3980–3980. Navigation pointer only: the counted text refers to the Stacks item lemma-regular-gorenstein, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original reference more-algebra-section-properties-formal-fibres: sources/stacks-dualizing/pinned-original-source.tex, source lines 4385–4385. Navigation pointer only: the counted text refers to the Stacks item more-algebra-section-properties-formal-fibres, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Properties (A), (B), (C), (D), and (E) of More on Algebra, Section more-algebra-section-properties-formal-fibres hold for $P(k \to R) =$"R is a Gorenstein ring".*

Proof. Since we already know the result holds for Cohen-Macaulay instead of Gorenstein, we may in each step assume the ring we have is Cohen-Macaulay. This is not particularly helpful for the proof, but psychologically may be useful.

Part (A). Let K/k be a finitely generated field extension. Let R be a Gorenstein k-algebra. We can find a global complete intersection $A = k[x_1, \ldots, x_n]/(f_1, \ldots, f_c)$ over k such that K is isomorphic to the fraction field of A, see Algebra, Lemma algebra-lemma-colimit-syntomic. Then $R \to R \otimes_k A$ is a relative global complete intersection. Hence $R \otimes_k A$ is Gorenstein by Lemma lemma-gorenstein-lci. Thus $R \otimes_k K$ is too as a localization.

Proof of (B). This is clear because a ring is Gorenstein if and only if all of its local rings are Gorenstein.

Part (C). Let $A \to B \to C$ be flat maps of Noetherian rings. Assume the fibres of $A \to B$ are Gorenstein and $B \to C$ is regular. We have to show the fibres of $A \to C$ are Gorenstein. Clearly, we may assume $A = k$ is a field. Then we may assume that $B \to C$ is a regular local homomorphism of Noetherian local rings. Then B is Gorenstein and $C/\mathfrak{m}_B C$ is regular, in particular Gorenstein (Lemma lemma-regular-gorenstein). Then C is Gorenstein by Lemma lemma-flat-under-gorenstein.

Part (D). This follows from Lemma lemma-flat-under-gorenstein. Part (E) is immediate as the condition does not refer to the ground field. □